AJAN MUTHURAJ

ROBOTICS SYSTEMS ENGINEER

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01 PROFILE

Systems Engineer at Dai-ichi Life Techno Cross in Japan and IIT Madras Mechanical Engineering graduate building end-to-end robotic systems across safe reinforcement learning, aerial autonomy, multi-robot mapping, motion planning, perception and manipulation. Strongest at turning research ideas into observable, reproducible systems that connect physical constraints, sensing, algorithms and deployment.

02 EDUCATION

Indian Institute of Technology Madras — 2022 - 2026

B.Tech, Mechanical Engineering | CGPA: 8.52/10

Thesis: Reach-Avoid Reinforcement Learning with Time-Varying Safe and Target Zones.

03 EXPERIENCE & RESEARCH

Systems Engineer - Dai-ichi Life Techno Cross — 2026 - Present

Japan

- Working in systems engineering following graduation from IIT Madras.

Robotics Research Intern - IIT Roorkee — Jun - Jul 2025

Humanoid navigation, perception and manipulation

- Integrated Gazebo/URDF simulation, SLAM localization and Nav2 autonomous navigation for a hospital-like environment.
- Built YOLOv8 and depth-based 3D localization, then connected MoveIt 2 and OctoMap for collision-aware manipulation; reported approximately 96.6% grasp success across 30 trials.

04 SELECTED PROJECTS

Reach-Avoid RL in Time-Varying Environments — Nov 2025 - May 2026

B.Tech thesis | PyTorch, DDQN, Gym

- Extended the discounted reach-avoid formulation with time-augmented state, corrected baseline implementation issues and trained a DDQN for moving obstacles and targets; reported up to 94.4% safe-state prediction precision.

Autonomous UAV Exploration in Isaac Sim — 2026

Isaac Sim, ROS 2, ArduPilot, RTAB-Map

- Built and documented a reproducible stack from simulated sensors and flight control through SLAM, collision-aware 3D frontier exploration and multi-drone launch workflows.

Multi-Robot Exploration & Temporal Map Fusion — 2025 - 2026

Collaborative research | SLAM, map fusion, control

- Worked on shared occupancy-grid memory, deadlock recovery and coordinated frontier exploration under noisy sensing; wider work accepted as an ICRA 2026 XPLORE Workshop poster.

Dynamic Path Planning — Apr 2025

Adaptive RRT*, LPA*, ROS 2, Gazebo

- Implemented and benchmarked planners in moving-obstacle scenarios; reported approximately 12% shorter paths on average in the evaluated cases.

05 TECHNICAL SKILLS

ROBOTICS	ROS 2, Nav2, MoveIt 2, Gazebo, Isaac Sim, RViz, ArduPilot, RTAB-Map, SLAM Toolbox
AI / PLANNING	PyTorch, DDQN, TD3, YOLOv8, OpenCV, RRT*, LPA*, frontier exploration
ENGINEERING	Python, C++, MATLAB, Linux, Git, Docker, Arduino, ESP32, system integration

06 RECOGNITION & CREDENTIALS

ICRA 2026 XPLORE Workshop: Poster acceptance for collaborative work on multi-robot frontier-based exploration under uncertainty with SLAM and temporal map fusion.
Certifications: Aerial Robotics - University of Pennsylvania (2025); Robotics Software Engineer Nanodegree - Udacity (2024); e-Yantra Robotics Competition - IIT Bombay (2024).